

Lachlan Watts-Tobin

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EDUCATION

University of California, Berkeley

B.S Electrical Engineering & Computer Science, Minor in Mechanical Engineering

Berkeley, CA

May 2028

SKILLS

Technical Skills: Computer-Aided Design (CAD), Finite Element Analysis (FEA), Data Entry, Computer Programming, 3D Printing, Metal Fabrication, Industrial Safety, Risk Reduction, Sustainable Design, Additive Manufacturing, Welding, Project Management.

Platforms: Solid Works, Autodesk Fusion 360, Python, MATLAB, Scheme, SQL, Java, RISC-V, C, C++, Adobe Creative Cloud, Microsoft Office 360 (Word, Excel, PowerPoint), Canva, Google Workspaces (Docs, Sheets, Slides).

EXPERIENCE

VISION IV

(Remote) Yokohama, Japan

Engineering Intern

May 2025 – July 2025

- Designed and optimized a high-performance component using experimental materials for reliability.
- Analyzed potential failure points to guide data-driven improvements to component performance.
- Developed a business plan covering strategy, resources, suppliers, and marketing.

Engineering Student Services - PREP Program

Berkeley, CA

Mathematics Learning Facilitator

June 2025 – August 2025

- Taught pre-calculus to ~100 incoming engineering students in a collaborative learning environment.
- Developed a rigorous pre-calculus curriculum with original lessons and assignments.
- Improved engagement by adapting instruction based on student and peer feedback.

Watts Electronics

Arcata, CA

Co-Founder & Chief Electrical Engineering

March 2026 – Present

- Design analog circuits for guitar effects pedals, inspired by vintage hardwired models.
- Manage small-scale development of units and handled sale to local music stores.

PROJECTS

Bicycle Telemetry System

Bike Builders of Berkeley, (January 2026 – March 2025)

- Built a system that gathers stress & strain on different sections of a bike while it's ridden, then saves data locally.

Scheme Interpreter

UC Berkeley, College of Engineering, (April 2025 - April 2025)

- Built a fully functional Scheme interpreter in Python, implementing parsing, evaluation, and optimization.

CS61CPU

UC Berkeley, College of Engineering, (November 2025 – December 2025)

- Designed a functional CPU in Logisim capable of executing 32-bit RISC-V instructions.

Carbon Fiber Filament Winder

Bike Builders of Berkeley, (June 2025 – December 2025)

- Designed and built a fully functional 3-axis carbon fiber filament winder to produce bike tube sections.

Algorithmically Generated G-Code Slicer & Controller

UC Berkeley, College of Engineering, (January 2025 – April 2025)

- Developed software to generate G-code from part file parameters and control a filament winder to create it.

LEADERSHIP

Bike Builders of Berkeley

Berkeley, CA

CFRP (Carbon Fiber Reinforced Polymer) Engineer

August 2024 – Present

- Continuously develop new techniques for vacuuming and post-processing, which drastically improve quality.
- Presented weekly updates on workflow, testing, and proposed process improvements to the entire team.

Cal Solar Vehicle

Berkeley, CA

Electrical Engineer

Jan 2026 – Present

- Designed a PCB speed controller for the primary motor and wrote corresponding firmware on the control board.